Toronto Metropolitan University

Metropolitan Undergraduate Engineering Society

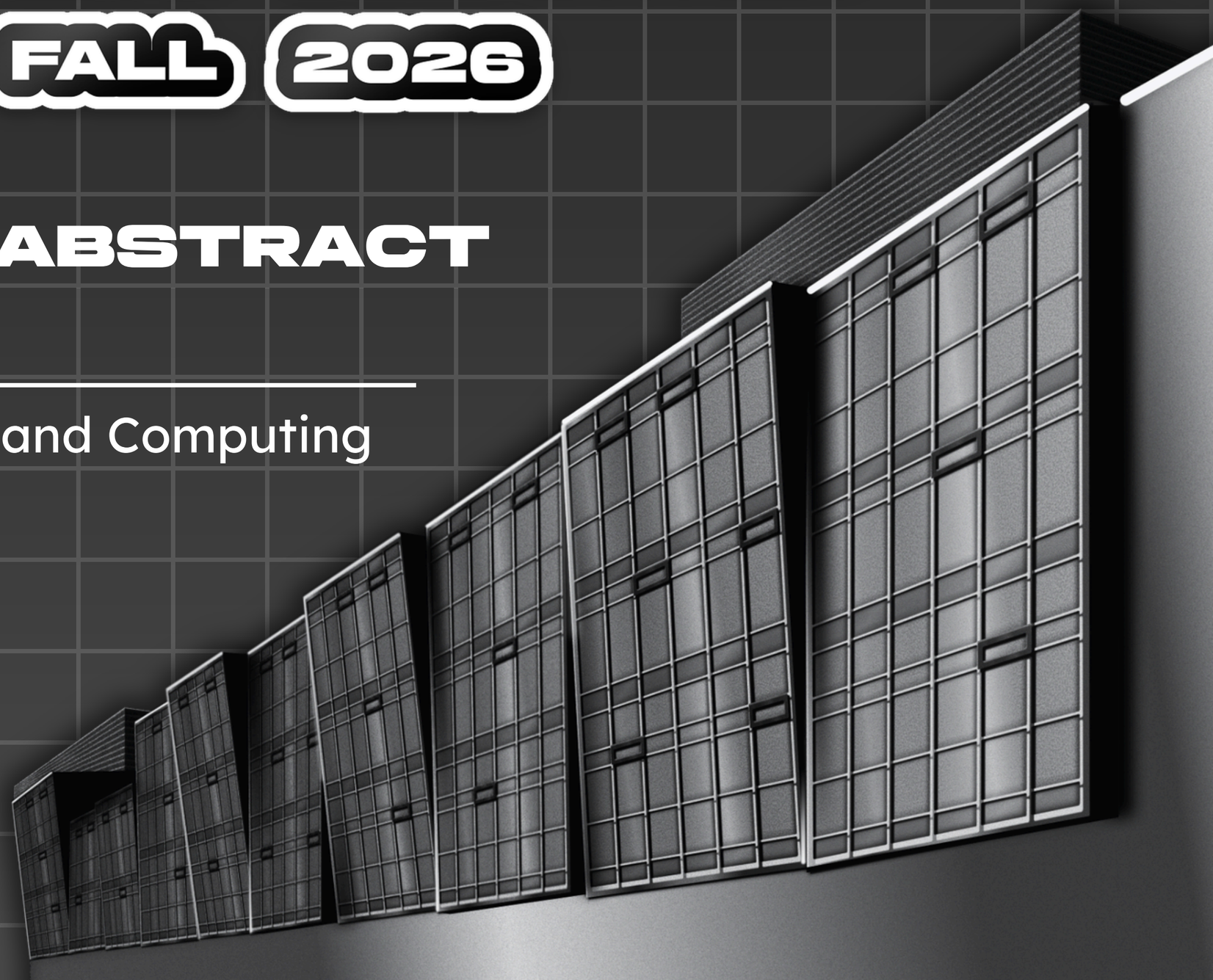
# TORONTO METROPOLITAN UNIVERSITY ENGINEERING COMPETITION

# FALL 2026

## CONSULTING ABSTRACT

NOVEMBER 7 & 8, 2026

George Vari Engineering and Computing Centre



## Competition Information

**Competition Date:** Saturday, November 7, 2026

**Team Size:** 3-4 competitors

**Working Time:** 8 hours

The official case package will be provided at the beginning of the competition. Teams must complete their consulting analysis, prepare their report and presentation materials, and submit the required deliverables by the deadline announced by the Competition Coordinators.

# 1. INTRODUCTION

Food security is an increasingly important challenge for growing urban communities. As cities expand, ensuring reliable access to affordable, nutritious, and culturally appropriate food requires careful consideration of infrastructure, transportation, sustainability, technology, and public resources.

The Metropolitan Engineering Competition (MEC) 2026 explores this challenge through the theme “Feeding the Future: Designing Smart Infrastructure for Urban Food Security.” This theme encourages competitors to examine how engineering can improve the accessibility, efficiency, and resilience of urban food systems while addressing the needs of diverse communities.

In the Consulting Engineering Competition, teams will take on the role of engineering consulting firms tasked with developing a practical, long-term infrastructure strategy for the City of Toronto.

Unlike a traditional design competition focused on a single product or prototype, this competition challenges participants to evaluate an interconnected system. Teams must consider how infrastructure, food storage, transportation, digital technologies, financial resources, and community needs can work together to support a more effective urban food network.

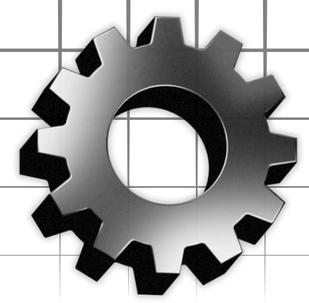
Successful proposals must demonstrate sound engineering judgment, realistic planning, and the ability to justify decisions under competing constraints.

# 2. BACKGROUND

## 2.1 Toronto's Urban Food System

Toronto's food system depends on a complex network of food producers, suppliers, distributors, retailers, community organizations, and public services.

Although these organizations contribute to food availability across the city, access to affordable and nutritious food is not equally distributed.



Some communities face greater challenges involving food affordability, transportation, access to appropriate food options, and the availability of nearby distribution services.

At the same time, organizations responsible for food distribution must manage several operational challenges, including:

- Food spoilage and waste.
- Transportation and distribution costs.
- Refrigeration and safe food storage.
- Limited land and infrastructure availability.
- Unpredictable demand.
- Supply-chain disruptions.
- Environmental impacts and energy consumption.
- Accessibility for communities with different dietary, cultural, medical, and mobility needs.

These challenges are interconnected. Improving one aspect of the food system may introduce additional costs or limitations elsewhere.

For example, increasing the number of distribution facilities may improve geographic accessibility but also increase construction, maintenance, and operating expenses.

Similarly, concentrating food storage and distribution into fewer large facilities may improve certain operational efficiencies while creating longer transportation routes or additional vulnerabilities during disruptions.

The City of Toronto must therefore consider how to balance accessibility, financial responsibility, operational efficiency, resilience, and sustainability.

## 2.2 The Client

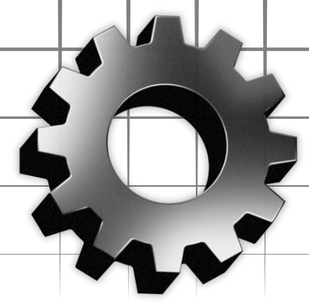
For this competition, teams will work with a **fictional municipal client**:

**City of Toronto - Food Security & Urban Infrastructure Division**

The division is seeking engineering consulting services to support the development of a coordinated food infrastructure network.

The client is interested in a long-term strategy that improves access to food while reducing waste, strengthening distribution capabilities, and making responsible use of municipal resources.

The client recognizes that no single approach will perfectly satisfy every objective.



Consulting teams are expected to evaluate alternatives, identify limitations, and recommend a defensible strategy.

## 3. PROPOSAL

In the MEC 2026 Consulting Engineering Competition, teams will act as engineering consulting firms advising the City of Toronto's simulated Food Security & Urban Infrastructure Division.

Your task is to develop a long-term Smart Food Hub Network for Toronto that improves food accessibility, reduces food waste, strengthens supply-chain resilience, and supports sustainable urban infrastructure.

The proposed Smart Food Hub Network must include a practical strategy for maintaining essential food distribution during a continuous 14-day supply-chain disruption.

Teams must identify the disruption assumptions used in their analysis, explain which parts of the proposed network remain operational, identify the resources required to maintain essential services, and justify the feasibility of their contingency strategy.

Judges will evaluate the quality and practicality of the proposed response using the same engineering feasibility and risk-analysis criteria for every team.

Teams must evaluate potential infrastructure strategies and recommend a practical solution that balances population coverage, cost, operational efficiency, land use, environmental sustainability, and the needs of diverse communities.

Competitors must analyze the provided information, identify the key engineering challenges, and evaluate potential solutions before selecting an approach that best addresses the client's needs. Teams are expected to consider technical feasibility, financial limitations, environmental sustainability, accessibility, and long-term operational requirements when developing their recommendations.

The proposed solution must be supported by appropriate engineering analysis, reasonable assumptions, and clear justification of important decisions and trade-offs. Teams should also demonstrate how their recommendations could be implemented in practice while considering relevant risks, resource requirements, and stakeholder impacts.

Teams may use reasonable and clearly stated assumptions where specific data is not provided. All assumptions must be justified and applied consistently throughout the proposed solution.

## 3.1 Mandatory Engineering Constraints

The proposed Smart Food Hub Network must satisfy the following requirements:

1. **Budget:** The total proposed cost of the Smart Food Hub Network must not exceed $200 million CAD over the 10-year planning period. This budget must account for relevant infrastructure development, equipment, implementation, transportation, operations, maintenance, and other costs necessary to support the proposed network.
2. **Population Coverage:** The proposed network must target service coverage for 85-95% of Toronto's population. Teams must clearly define how they measure service coverage, such as geographic accessibility, reasonable travel distance, or access through delivery services. Teams must justify their chosen measurement approach and explain how the remaining population will be supported through alternative or supplementary food-access strategies.
3. **Supply-Chain Resilience:** The network must include a practical strategy for maintaining essential food distribution during a 14-day supply-chain disruption.
4. **Food-Waste Reduction:** Teams must propose engineering and operational measures targeting a 30-40% reduction in food waste compared with a clearly defined baseline. Teams must explain their assumptions, identify the sources of food waste being addressed, and describe how the expected improvements would be measured.

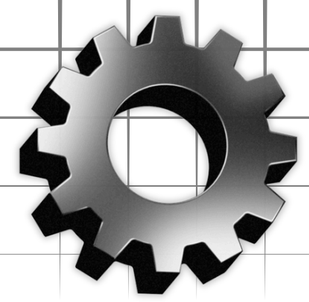
   Teams must establish and justify an appropriate baseline for food waste within the scope of their proposed network.

   The expected percentage reduction must be supported by reasonable assumptions, relevant data, or engineering calculations. Teams should explain which operational improvements contribute to the reduction and distinguish estimated outcomes from verified results.
5. **Land and Infrastructure:** Proposed facilities must consider land availability, zoning, transportation access, storage capacity, and long-term infrastructure requirements.
6. **Accessibility and Equity:** Teams must consider the needs of underserved communities, including dietary, cultural, medical, and mobility-related accessibility requirements.

Teams must justify their proposed strategies using appropriate engineering analysis, financial estimates, reasonable assumptions, and supporting evidence.

## 3.2 Budget Allocation Guidance

The City of Toronto has established a maximum budget of $200 million CAD over 10 years.



The following allocation is provided as a suggested starting point:

| Category | Suggested Allocation |
|---|---|
| Infrastructure Construction | 40% — $80 million |
| Refrigeration and Food Storage | 20% — $40 million |
| Smart Sensors and Digital Infrastructure | 15% — $30 million |
| Transportation and Logistics | 15% — $30 million |
| Emergency Reserve and Maintenance | 10% — $20 million |
| Total | 100% — $200 million |

Teams may introduce additional budget categories, including staffing and ongoing operating expenses, where necessary to support their proposed network, provided the total 10-year budget remains within $200 million CAD.

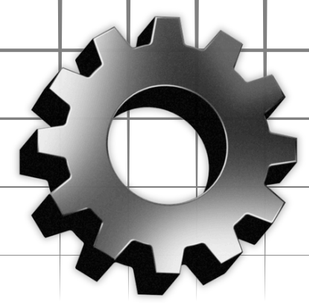
Teams may modify this allocation to support their proposed network, provided they remain within the overall budget and justify significant changes.

The budget should account for relevant infrastructure, equipment, implementation, maintenance, and operating requirements over the proposed planning period.

At the end of the competition, each team will present its findings through a professional consulting report and a presentation to a panel of judges acting as client representatives. Teams will be evaluated on the quality of their analysis, practicality of their recommendations, and ability to communicate and defend their engineering decisions.

## 4. DELIVERABLES

Each team must prepare a written consulting report and presentation slides. Both deliverables must be submitted in PDF format by the official competition deadline communicated during the competition briefing.



Teams will present their recommendations to a panel of judges acting as representatives of the client.

Both deliverables must communicate the team's proposed recommendation clearly and demonstrate the reasoning behind its engineering decisions.

### 4.1 Written Report

Teams must submit a professional engineering consulting report documenting their analysis and proposed recommendation.

**Report Requirements:**

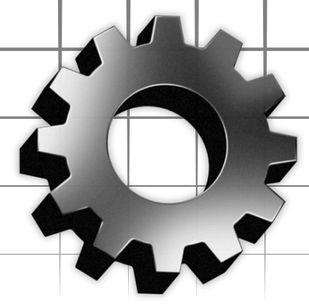
The final consulting report must not exceed ten (10) pages of main content, excluding the title page and references. Appendices may contain brief supporting information but must not be used to bypass the ten-page limit.

The report must include:

1. **Problem and Client Analysis:** Identify the key engineering challenges, client objectives, stakeholder needs, and limitations of Toronto's existing food infrastructure.
2. **Proposed Smart Food Hub Network:** Explain the recommended infrastructure strategy, including the proposed facilities, distribution approach, and relevant engineering systems.
3. **Engineering Representation:** Include at least one network diagram, infrastructure layout, or system representation illustrating how the proposed network would operate.
4. **Financial Analysis:** Provide an itemized budget and justify the major infrastructure, equipment, transportation, operating, and maintenance costs.
5. **Population Coverage and Equity:** Teams must clearly define their population coverage methodology and support their estimated coverage using appropriate data, calculations, maps, or documented assumptions.

   The proposed coverage calculation should identify the population considered accessible to the network, the criteria used to determine accessibility, and any communities that remain underserved.

   Unsupported claims of achieving the 85-95% coverage target will not be considered sufficient engineering evidence.
6. **Engineering Trade-Offs:** Identify and justify at least three major trade-offs affecting the proposed strategy.
7. **Implementation Roadmap:** Provide a realistic 10-year implementation strategy outlining major phases, infrastructure development, and operational priorities.
8. **Risk and Resilience Analysis:** Identify at least three major risks with mitigation strategies,



including a plan for maintaining essential operations during a 14-day supply-chain disruption.

9. **Sustainability and Food-Waste Reduction:** Explain how the proposed network supports the 30-40% food-waste reduction target and considers long-term environmental and operational sustainability.
10. **Engineering Justification:** Support significant decisions with appropriate calculations, evidence, assumptions, and technical reasoning.

All external sources must be appropriately cited. Appendices may be included for concise supporting information but must not be used to bypass the report's page limit.

The report must be professionally organized and written for an engineering consulting client.

### 4.2 Presentation

Each team must present its consulting recommendation to a panel of judges acting as representatives of the client.

**Presentation Duration:** 10 minutes
**Question-and-Answer Period:** 5 minutes

The presentation should summarize the team's understanding of the problem, proposed recommendation, supporting engineering analysis, and justification for its major decisions.

Teams should communicate their findings clearly and professionally, using appropriate visual materials to support their explanations.

During the question-and-answer period, judges may ask competitors to clarify their assumptions, explain technical decisions, defend trade-offs, or address concerns regarding their proposed recommendation.

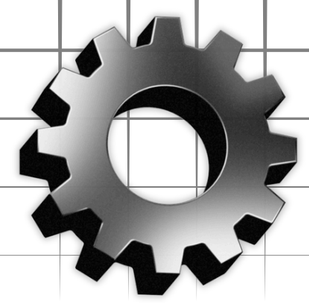
All team members are expected to contribute meaningfully to the competition.

## 5. GUIDELINES

### 5.1 Allowed Resources

Competitors may use the following resources during the competition, subject to MEC's official rules:

- The case materials supplied by organizers.
- Personal laptops and permitted electronic devices.
- Engineering software and calculation tools.
- Spreadsheet, presentation, and document software.



- Publicly available technical references and research materials.
- Other resources explicitly authorized by competition organizers.

Teams are responsible for ensuring that external information is relevant, accurate, and appropriately cited.

Any use of artificial intelligence must comply with MEC's official AI policy. Competitors remain responsible for the originality, accuracy, and technical justification of their submissions.

### 5.2 Question Periods

Competitors may seek clarification from competition organizers during designated question periods.

Questions should concern the interpretation of the competition requirements, provided case information, or applicable rules.

Organizers may decline to answer questions that would provide an unfair advantage or directly solve part of the engineering challenge.

Any clarification affecting competition requirements will be communicated consistently to all participating teams.

The timing and procedure for question periods will be announced on competition day.

### 5.3 Competition Rules

The following rules outline the requirements for the Consulting Engineering Competition. Additional supporting information, technical data, and scenario details may be provided in the official competition-day case package. Unless explicitly amended by the Competition Organizers, the requirements stated in this abstract remain applicable.

1. Teams must develop an original engineering consulting proposal addressing the assigned Smart Food Hub Network challenge.
2. All proposals must respect the mandatory budget, population coverage, resilience, and food-waste reduction requirements.
3. Teams must support major engineering and financial decisions with reasonable analysis, calculations, or justified assumptions.
4. The written report must not exceed 10 pages, excluding the title page and references.
5. Teams must include at least one engineering network or infrastructure representation.
6. Teams must identify at least three major trade-offs and three major risks with mitigation strategies.
7. Teams must submit the required report and presentation materials according to the deadlines

and procedures announced by organizers.

8. All external sources must be appropriately cited. Plagiarism and academic misconduct are prohibited.
9. Competitors must follow MEC's official conduct, confidentiality, and AI-use policies.
10. Teams must follow all additional instructions communicated by competition organizers.

## 6. JUDGING CRITERIA

The competition will be evaluated using the following categories:

| Judging Category | Points |
|---|---|
| Solution | 60 |
| Problem Understanding & Analysis | 15 |
| Engineering Feasibility & Implementation | 15 |
| Equity & Accessibility | 10 |
| Trade-offs & Risk Analysis | 10 |
| Long-Term Sustainability & Innovation | 10 |
| Report | 15 |
| Organization & Professionalism | 5 |
| Technical Evidence & Supporting Analysis | 5 |
| Recommendation & Design Justification | 5 |
| Presentation | 25 |
| Delivery & Professional Communication | 10 |
| Organization & Visual Communication | 5 |
| Technical Understanding & Q&A | 5 |
| Team Participation & Time Management | 5 |

| | |
|---|---|
| TOTAL | 100 |